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The Impact of Digital Economy Development on Regional Income Gaps: A Perspective on Multidimensional Inequality Decomposition and Threshold Effects

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Abstract: This paper investigates the impact of digital economy development on regional income gaps within provinces in China, focusing on the underlying mechanisms and multidimensional threshold effects. Using provincial panel data from 2011 to 2021, this study finds that the relationship between digital economy development and income gaps follows a U-shape, initially reducing the gap and later widening it. This analysis identifies internet penetration and digital transaction growth as key contributors to multidimensional inequality in digital economy development. Moreover, moderate levels of multidimensional inequality in digital economy development enhance its positive effect on narrowing regional income gaps, while excessive multidimensional inequality may diminish or even reverse this effect. These findings highlight the importance of managing multidimensional inequality to maximize digital economies' potential for fostering more balanced regional development.

Keywords: digital economy; regional income gaps; multidimensional inequality decomposition; common prosperity; threshold effect

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1. Introduction

Digital economies have become a driving force in the economic development of many countries, with their integration across various sectors of the economy and society deepening continuously. It plays a pivotal role in stimulating consumption, driving investment, and creating employment. Following the adoption of the G20 Digital Economy Development and Cooperation Initiative at the 2016 G20 Hangzhou Summit, digital economies emerged as a key topic for discussion among member countries and partner organizations. This initiative encourages the global development of digital economies, aiming to ensure that their benefits reach people worldwide. Subsequent G20 summits—held in Germany in 2017, Argentina in 2018, Japan in 2019, and Saudi Arabia in 2020—reaffirmed the digital economy's role as a major driver of global economic growth. At the 3rd BRICS Communications Ministers' Meeting, the BRICS nations emphasized innovation, interconnectivity, and the strengthening of information exchange and digital connectivity. The BRICS countries called for closer collaboration and the expansion of digital technologies across industries. The BRICS Leaders' Xiamen Declaration also advocated for creating conducive conditions for the flourishing of digital economies, ensuring their growth benefits are widely shared. According to the 2024 Global Digital Economy White Paper by the China Academy of Information and Communications Technology, digital economies continue to grow rapidly

on a global scale. By 2023, the combined digital economies of the U.S., China, Germany, Japan, and South Korea had exceeded USD 33 trillion, accounting for 60% of their GDP.

From a domestic perspective of China, since the 18th National Congress of the Communist Party, China has successively issued digital economy development strategies and the "14th Five-Year Plan" for the development of a digital economy, which have propelled the sector's robust growth. According to the China Academy of Information and Communications Technology (CAICT), the scale of China's digital economy has grown from CNY 11.2 trillion in 2012 to CNY 53.9 trillion in 2023, representing a 3.8-fold expansion over the 11-year period. From an international perspective, the "2024 Global Digital Economy White Paper" released by the CAICT indicates that digital economies continue to expand rapidly worldwide. In 2023, the combined digital economies of the United States, China, Germany, Japan, and South Korea exceeded USD 33 trillion, with digital economies accounting for 60% of GDP. While maintaining high-speed growth, the digital economy has gradually become an essential component and driving force of China's national economic development.

Meanwhile, over the past 40 years since the reform and opening up, China's economy has achieved remarkable high-speed growth, but significant disparities in economic development within provinces persist. Taking Guangdong as an example, in 2022, the GDP of Shenzhen reached CNY 3.24 trillion, while Yunfu, at the bottom of the rankings, had only CNY 116.2 billion, a difference of approximately 28 times (data obtained from the "Guangdong Statistical Yearbook 2023"). Following China's transition to the new normal, the issue of internal development imbalances among provinces has drawn increasing attention from the government (The regional income gaps discussed in this paper refer to the income gap between regions within each province.). The "Recommendations of the Central Committee of the Communist Party of China on the 13th Five-Year Plan for National Economic and Social Development" explicitly proposed that, while maintaining moderate-to-high economic growth, China should focus on enhancing the balance, inclusiveness, and sustainability of development, especially by reducing local policy barriers and promoting fair competition to alleviate internal development imbalances among provinces.

Although numerous measures have been taken by the government over the years to reduce the development gaps within provinces, the issue of regional income gaps remains unresolved and has become a major factor influencing the overall income inequality in China (Zhou et al. [1]). The severe regional income gaps, on one hand, suppress consumption demand, thereby restricting the sustainable growth of China's macroeconomy (Zhou and Ni [2]), while on the other hand, they create a sense of relative deprivation among low- and middle-income groups that is greater than the income gap between provinces (Feng [3]; Ma et al. [4]), which negatively impacts long-term social stability.

In this context, the digital economy, with its unique advantages in promoting efficient resource allocation and inter-regional factor mobility, is seen as a key solution to the problem of imbalanced and insufficient economic development (Duan et al. [5]). How to fully leverage the development dividends brought by a digital economy to address intra-provincial regional income gaps, achieve both regional economic growth and coordinated development, and ensure that all people share the benefits of reform and development, has become a critical issue in driving China's realization of common prosperity and the construction of the new "dual circulation" development pattern. Indeed, the existing literature has long discussed the issue of income gaps, but through a review of relevant studies, this paper finds that research on the impact of digital economies on income gaps has mostly focused on two aspects: income inequality among residents and the income gap between urban and rural areas.

Specifically, the first group of literature focuses on how a digital economy affects the income gap among residents. For example, Luo and Wang [6] uses data from the China Family Panel Studies (CFPS) and finds that the development of a digital economy enhances labor income, especially for low-income groups, suggesting that a digital economy has a certain level of inclusiveness and helps narrow the income gap among workers. At the

same time, Zhao and Peng [7] conducts an empirical analysis using provincial panel data from 2013 to 2019 and applies a spatial econometric model to study the impact of a digital economy on income inequality among residents. The results show that, to some extent, the development of a digital economy deepens income inequality and further expands income gaps in surrounding areas through spatial spillover effects.

The second group of literature examines how a digital economy affects the urban–rural income gap. The research by Chen and Wu [8] suggests that in the early stages of digital economy development, improving its growth level can effectively curb the widening urban–rural income gap. However, when a digital economy reaches a certain level, further improvement may not only fail to narrow the income gap but may even exacerbate income inequality between urban and rural areas. Wang and Xiao [9] also verifies that the impact of a digital economy on the urban–rural income gap follows a positive U-shape and has significant spatial spillover effects. Li and Li [10] further points out that the influence of digital economy development on the urban–rural income gap exhibits a threshold effect, where in regions with higher per capita income and greater R&D intensity, the positive effect of a digital economy on narrowing the urban–rural income gap is more pronounced. In contrast, the studies by Luo et al. [11] and Cheng and Zhang [12] show that the impact of a digital economy on the urban–rural income gap follows an inverted U-shape, where in the early stages of development, it widens the urban–rural income gap, but in the later stages, it gradually reduces the gap.

Undoubtedly, existing research has provided valuable empirical insights into the impact of the digital economy on income gaps in China. However, most studies have focused on a single dimension of digital economy development, such as internet penetration or mobile communication spread, and have rarely analyzed its overall impact from a multidimensional perspective. This single-dimensional framework is limited in revealing the complex mechanisms of a digital economy's effects. Furthermore, the literature lacks in-depth discussions on the pathways through which a digital economy influences regional income gaps within provinces, especially the threshold effect of multidimensional inequality of digital economy development, which has yet to be fully explored.

Therefore, this paper starts from the essential characteristics of a digital economy and conducts a theoretical analysis of how it influences regional income gaps within provinces, focusing on the mechanisms through which it operates and its threshold effects based on its multidimensional development. Through an empirical study of provincial panel data from 2011 to 2021, the results show that the development of a digital economy exhibits a positive U-shaped nonlinear impact on regional income gaps. Further, by calculating the multidimensional inequality index of digital economy development and its decomposition, and constructing a threshold regression model, this paper finds that within a specific range of multidimensional inequality, the positive impact of digital economy development on reducing regional income gaps is particularly significant. Moderate inequality levels are more effective in unleashing the potential of a digital economy to narrow regional income gaps.

This paper makes several contributions to the literature. First, it enhances the understanding of how digital economy development influences regional income gaps, offering new insights into the relationship between technological development and income distribution. Second, it develops a measurement framework for digital economy development using machine learning algorithms—Random Forest, GBDT, and XGBoost—applied at the city level. Third, it introduces a methodological innovation by using the Kmenta approximation to decompose the multidimensional inequality of digital economy development, isolating the contributions of each dimension.

2. Theoretical Analysis and Research Hypotheses

The core of high-quality development lies in achieving balanced income distribution, rational resource allocation, and coordinated economic growth within regions. A digital economy, with its advantages in cross-temporal and cross-spatial information dis-

semination, data generation and sharing, and significant reductions in transaction costs, effectively alleviates the conflicts in factor supply and demand, spatial limitations of economic activities, and the difficulty of balancing fairness and efficiency in regionally balanced development. This provides a new feasible path to reduce regional income gaps. In addition to the direct impact of a digital economy's characteristics on regional income gaps, its development involves collaborative progress across multiple dimensions, where the balance of development in each dimension may result in different threshold effects on income gaps. Based on this, this paper examines and demonstrates the influence of a digital economy on intra-provincial income gaps from the perspectives of its mechanisms and the threshold effects of multidimensional synergies, and subsequently proposes the relevant research hypotheses.

2.1. The Mechanisms of Digital Economies' Impact on Regional Income Gaps

2.1.1. Inclusive Phase: Narrowing Gaps

In its early stages, a digital economy often demonstrates strong inclusivity, promoting balanced development among regions within a province. By improving production efficiency and reducing transaction costs, a digital economy provides underdeveloped regions with increased access to markets and resources. This access helps these regions participate more effectively in economic activities, thereby narrowing income gaps.

Specifically, the development of inclusive digital infrastructure, the expansion of e-commerce, and the proliferation of remote services create valuable opportunities for economic participation in resource-scarce regions. For instance, the widespread adoption of e-commerce platforms dismantles geographical and informational barriers in traditional economic activities, allowing small businesses and individual vendors to reach broader markets, thereby driving significant income growth in underdeveloped regions. Additionally, the promotion of digital services such as telemedicine and online education substantially enhances productivity and fosters the accumulation of human capital in these regions, contributing to a more equitable income distribution. Hence, during the inclusive phase, a digital economy effectively narrows income gaps between underdeveloped and developed regions within provinces by diffusing knowledge and technology, showcasing its pronounced gap-narrowing effect.

2.1.2. Capital-Intensive Phase: Widening Gaps

As a digital economy enters a more mature stage of development, its growth model increasingly exhibits capital-intensive characteristics, relying heavily on advanced technological innovation, capital investment, and the concentration of human resources, which are typically concentrated in more-developed regions.

During this phase, the development of a digital economy becomes increasingly shaped by the "Matthew effect," where resource-rich and developed regions capitalize on their accumulated advantages in capital, technology, and human resources to secure a disproportionate share of the growth dividends, thereby reinforcing their income dominance. Research highlights that technologies such as artificial intelligence intensify income inequality by increasing the demand for high-skilled workers while diminishing opportunities for low-skilled labor ([13]). These developed regions not only gain accelerated access to advanced technological resources but also attract an influx of highly skilled talent and capital investments, spurring rapid income growth and deepening economic disparities within regions.

In contrast, underdeveloped regions, lacking the necessary capital, technology, and human resource support, struggle to compete effectively, gradually falling behind in the digital economic development process. This results in a widening income gap within provinces (Piketty [14]; Acemoglu and Restrepo [15]).

2.1.3. Nonlinear Trend: U-Shaped Effect

Building on the above analysis, the impact of a digital economy on regional income gaps within provinces exhibits a nonlinear trend of "narrowing-then-widening". During the "inclusive" phase, a digital economy, with its low entry barriers and knowledge diffusion effects, promotes the development of underdeveloped regions, thereby reducing the income gap within provinces. However, as a digital economy gradually enters the "capital-intensive" phase, resources increasingly concentrate in developed regions, leading to an expansion of the income gap. This shift reveals that the impact of a digital economy on the provincial income gap is not linear, but dynamically adjusts according to the stage of development. Based on this analysis, the following hypothesis is proposed.

H1: The impact of digital economy development on regional income gaps within provinces follows a positive U-shaped relationship, initially narrowing and then widening the gap.

2.2. Multidimensional Threshold Effects of Digital Economies' Impact on Regional Income Gaps

The development of a digital economy has distinct multidimensional characteristics, and the overall level of multidimensional inequality, as well as the inequality within each dimension, has a multilayered impact on regional income gaps within provinces (Goldfarb and Tucker [16]). Within the framework of this paper, the multidimensional structure of a digital economy—such as internet penetration, information technology talent, communication service development, mobile communication penetration, and digital transaction development—creates varying impact paths on income growth, depending on the regional development gaps.

2.2.1. Positive Effects of Balanced Multidimensional Development in a Digital Economy

When these dimensions are relatively balanced, underdeveloped regions are better able to absorb the technological spillover effects and market opportunities generated by a digital economy, thereby improving their productivity and income levels. In this context, the inclusive effects of a digital economy become more pronounced, helping to reduce regional income gaps.

For example, moderate internet penetration provides underdeveloped regions with access to markets and information, thus enhancing local economic vitality (Teece [17]; Van Deursen and Van Dijk [18]). Additionally, in situations of moderate inequality, digital transactions can promote the rational allocation of resources, preventing the excessive concentration of resources and technology, thereby balancing the development opportunities across regions (Philippon [19]).

2.2.2. Adverse Effects of Excessive Multidimensional Inequality in Digital Economy Development

However, when the overall multidimensional inequality of digital economy development, or the inequality within certain dimensions, exceeds a reasonable range, resources and technology may gradually tilt toward more-developed regions. Consequently, the participation of underdeveloped regions in the digital economy will decrease (Tan et al. [20]). As these regions struggle to meet the digital economy's demands for capital, technology, and human resources, they will gradually lose opportunities to benefit from the digital dividends, further exacerbating regional income gaps (Wang and Zhang [21]; Van Dijk and Hacker [22]).

High levels of inequality can also lead to the concentration of resources in a few economically advanced regions, intensifying regional differentiation and making it difficult for underdeveloped regions to enjoy the technological diffusion effects (Bloom et al. [23]; Forman et al. [24]). As a result, under these conditions, a digital economy may actually contribute to the widening of regional income gaps, creating a pronounced economic imbalance within provinces.

Therefore, the development inequality of a digital economy as a whole and across different dimensions should remain within a reasonable range between regions to maximize its gap-narrowing effect. When inequality, both overall and within each dimension, is controlled within reasonable bounds, a digital economy can more effectively exert its inclusive effect, reducing regional income gaps. However, when the level of inequality exceeds the reasonable range, the positive effect of a digital economy on underdeveloped regions will gradually weaken or even reverse, leading to a further widening of income gaps (Fan et al. [25]). Based on this, the following hypothesis is proposed in this paper.

H2: The impact of an overall digital economy and its various dimensions of development of inequality on regional income gaps within provinces follows a significant threshold effect. Specifically, when inequality levels remain within a reasonable range, a digital economy's ability to narrow regional income gaps is maximized; beyond this threshold, its positive effect diminishes or may even reverse.

3. Empirical Model Design, Index Construction, and Data Description

3.1. Empirical Model Design

Considering data availability and consistency in statistical standards, this paper employs panel data from 30 provinces over the period 2011–2021 to empirically examine the impact of digital economy development on regional income gaps. To test the above hypotheses, the following baseline model is constructed:

$$Gini_{it} = \alpha + \beta_1 DEI_{it} + \beta_2 DEI_{it}^2 + \gamma X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where $Gini_{it}$ denotes the regional income gaps within province i in year t; DEI_{it} represents the digital economy development index of province i in year t, reflecting the level of digital economy development in that province; the quadratic term DEI_{it}^2 is included to capture the nonlinear effects of digital economy development on regional income gaps. X_{it} is a vector of control variables, covering the following aspects: government intervention (gov), measured by the ratio of fiscal expenditure to regional GDP, which reflects the degree of government intervention and support in regional economic activities; research and development intensity (rd), represented by the ratio of internal R&D expenditure to regional GDP, indicating the level of investment in innovation and technological progress; economic development level ($gdpp$), measured by per capita GDP, reflecting the relative prosperity of the regional economy; the level of technology market development ($tech$), measured by the ratio of technology market transaction volume to GDP, assessing the region's activity in technology trading and transformation; and, finally, industrial structure ($stru$), represented by the ratio of tertiary industry output to secondary industry output, which reflects the modernization of the regional economic structure and the proportion of the service sector, illustrating the progress in economic structural transformation. Provincial fixed effects μ_i control for time-invariant characteristics among provinces, while year fixed effects λ_t control for the influence of macroeconomic fluctuations and policy changes, and ε_{it} is the random error term.

In the baseline model, β_1 and β_2 capture the linear and nonlinear effects of digital economy development on regional income gaps within provinces, respectively. Based on theoretical analysis, the expectation is that $\beta_1 < 0$, indicating that in the early stages of digital economy development, expansion of a digital economy can reduce regional income gaps, as the spread of digital technology often provides new market opportunities and means of economic participation for less-developed regions. However, it is expected that $\beta_2 > 0$, meaning that once digital economy development reaches a certain level of maturity, income gaps may widen again, showing a U-shaped nonlinear characteristic. This could occur because, at advanced stages of digital economy development, more-developed regions further benefit from the spillover dividends of a digital economy due to their stronger technological, capital, and human resource advantages, while less-developed regions, constrained by limited resources, struggle to keep pace, thereby widening the income gap.

To further verify the hypotheses proposed in the theoretical analysis, this paper introduces the multidimensional inequality index of digital economy development and its decomposed dimensions as threshold variables into the baseline model. By employing a panel threshold model, the impact of the interaction between these factors and the level of digital economy development on regional income gaps is analyzed. The specific models are constructed as follows:

$$\begin{aligned} Gini_{it} =& \alpha + \beta_1 DEI_{it} I(MGini_{it} \leq \tau_1) + \beta_2 DEI_{it} I(\tau_1 < MGini_{it} \leq \tau_2) \\ & \beta_3 DEI_{it} I(MGini_{it} > \tau_2) + \gamma X_{it} + \mu_j + \lambda_t + \varepsilon_{it} \end{aligned} \tag{2}$$

$$\begin{aligned} Gini_{it} =& \alpha + \beta_1 DEI_{it} I(MGini_{ikt} \leq \tau_1) + \beta_2 DEI_{it} I(\tau_1 < MGini_{ikt} \leq \tau_2) \\ & \beta_3 DEI_{it} I(MGini_{ikt} > \tau_2) + \gamma X_{it} + \mu_j + \lambda_t + \varepsilon_{it} \end{aligned} \tag{3}$$

In Equations (2) and (3), this paper uses the multidimensional inequality index of digital economy development, $MGini_{it}$, and its five decomposed dimensions, $MGini_{ikt}$ (where $k = 1, 2, \ldots, 5$), as threshold variables to examine the impact of digital economy development on regional income gaps at different levels of inequality. The first model employs the overall multidimensional inequality index $MGini_{it}$ of digital economy development, setting two threshold values, τ_1 and τ_2, to divide inequality levels into low, medium, and high intervals, thus identifying the varying impacts of digital economy development on regional income gaps across these levels. The second model further refines the inequality index to the level of specific dimensions, $MGini_{ikt}$, also divided into three intervals, to capture the distinct threshold effects of inequality levels in each specific dimension on the impact of digital economy development on regional income gaps. This enables a deeper exploration of the nonlinear characteristics of multidimensional inequality in digital economy development as it influences regional income gaps.

3.2. Index Construction and Data Description

3.2.1. Measurement of Regional Income Gaps

For measuring regional income gaps, this paper adopts the approach of Zhou et al. [26] and Liu et al. [27], using the average nighttime light intensity of each region as a proxy for income levels to assess income gaps across regions. Nighttime light data are chosen for their high spatial and temporal resolution, which captures economic activity distribution down to the city and even township levels. Compared with traditional income data, nighttime light data offer the advantages of wide geographic coverage and ease of access, particularly in underdeveloped regions where economic statistics may be incomplete or updated infrequently. These attributes enable nighttime light data to serve as a more stable and timely indicator of economic activity. Moreover, its strong correlation with indicators like per capita GDP has been validated in numerous studies, establishing it as a robust tool for measuring regional income gaps.

However, nighttime light data are not without limitations. First, light pollution in urban areas can overestimate economic activity, potentially introducing bias in densely populated regions. Second, in less-developed regions, the lower resolution of light data may obscure finer economic disparities. Lastly, variations in satellite calibration over time could affect data consistency, although preprocessing efforts help mitigate these effects.

Despite these challenges, the advantages of nighttime light data—their extensive spatial coverage, timeliness, and validated correlation with economic indicators—make them a reasonable and effective proxy for assessing regional income levels. Their ability to capture economic activity, even in regions with sparse traditional data, underscores their suitability for this study's analysis of regional income gaps. The nighttime light data used in this paper come from the NPP-VIIRS Nighttime Light (NTL) dataset, which provides global nighttime light intensity information, covering a long-term time series from 2000 to 2023 Chen et al. [28].

To measure regional income gaps within provinces, this paper primarily employs the Gini coefficient, which ranges from 0 to 1, with higher values indicating greater income

gap (while the Gini index is widely used to measure income inequality, it can be unreliable when applied to datasets with heavy-tailed distributions, particularly in the presence of aggregation and nonlinearity (Fontanari et al. [29]; Maia et al. [30]). Traditional income data often suffer from biases, especially due to the underrepresentation of high-income groups, which can lead to inaccurate estimates of income inequality. In contrast, nighttime light data, derived from satellite observations, offer broad spatial coverage and more comprehensive insight into overall economic activity. This approach helps mitigate biases commonly associated with traditional income surveys by circumventing issues related to missing high-income groups. By calculating the Gini index based on the annual average of nighttime light data at the city level and aggregating it to the provincial level, this study ensures that biases from missing high-income groups are minimized, enhancing the reliability of the estimated income gaps.). The calculation method is as follows:

$$Gini_i = \frac{\sum_{c=1}^{n_i} \sum_{c'=1}^{n_i} |y_c - y_{c'}|}{2n_i(n_i - 1)\mu_i} \tag{4}$$

In Equation (4), i represents the province, n_i is the number of cities within province i; y_c and $y_{c'}$ represent the average nighttime light intensity of cities c and c' within the province, respectively; μ_i is the average nighttime light intensity of all cities in province i, and $|y_c - y_{c'}|$ represents the absolute difference between cities. The denominator $2n_i(n_i - 1)$ is a normalization factor that confines the result to the range of 0 to 1.

3.2.2. Measurement of Digital Economy Development Level

Existing studies on the measurement of digital economy development levels often utilize provincial-scale data; however, provincial-level measurements can struggle to capture the detailed information available at the city level. In light of this, this paper draws on the approaches of Huang et al. [31] and Zhao et al. [32], focusing on internet development as a core measurement direction and incorporating indicators related to digital transactions. Based on available city-level data, the level of digital economy development is measured across five dimensions: internet penetration, information technology talent, communication service development, mobile communication penetration, and digital transaction development. These city-level indices are subsequently aggregated to derive the provincial-level digital economy development index. Specifically, the dimension of digital transaction development is measured using the China Digital Inclusive Finance Index Guo et al. [33]; Zhang et al. [34]. Internet penetration is represented by the number of internet broadband subscribers per 100 people, the information technology talent dimension by the proportion of employees in the computer services and software industry, communication service development by the per capita volume of telecommunications services, and mobile communication penetration by the number of mobile phone users per 100 people. The data primarily come from the annual "China City Statistical Yearbook", provincial "Statistical Yearbooks", and the Digital Finance Research Center at Peking University. For missing data in certain years, interpolation methods are applied.

Figure 1 presents the construction process of the digital economy development index. This process involves three main stages: normalizing raw data to ensure comparability, assigning indicator weights using machine learning algorithms, and aggregating weighted indicators into city- and provincial-level indices. These steps collectively enable the calculation of a comprehensive and robust index that reflects the level of digital economy development across regions.

The process begins with normalizing the raw data to a 0–1 scale, eliminating unit differences across indicators and ensuring comparability on a unified scale. Next, to determine the response variable values for the machine learning model, the K-means clustering algorithm is employed to naturally classify each city's digital economy development level, dividing them into distinct development groups. The optimal number of clusters is determined based on clustering evaluation metrics, including the Hopkins statistic, average silhouette coefficient, and Calinski–Harabasz index, ultimately producing clustering labels

for each city. These labels are used in the subsequent weighting stage of the machine learning model.

To assign weights to the indicators in the digital economy development index, this paper employs three ensemble learning algorithms: Random Forest, GBDT, and XGBoost. Each algorithm evaluates feature importance differently, capturing the nonlinear relationships between each city's digital economy development and its clustering label. Specifically, the Random Forest model assesses feature importance based on the decrease in Gini impurity, the GBDT model calculates feature weights through a reduction in deviation, and the XGBoost model evaluates feature importance using a reduction in cross-entropy. To enhance the robustness of weight calculations, a model-averaging approach is applied, taking the arithmetic mean of the weights from the three models as the final weight for each indicator, providing a more accurate reflection of each indicator's importance in digital economy development. Finally, based on the standardized data and composite weights, a linear weighting method is used to aggregate each city's digital economy development index, with the provincial-level index represented by the linear average of city indices within each province.

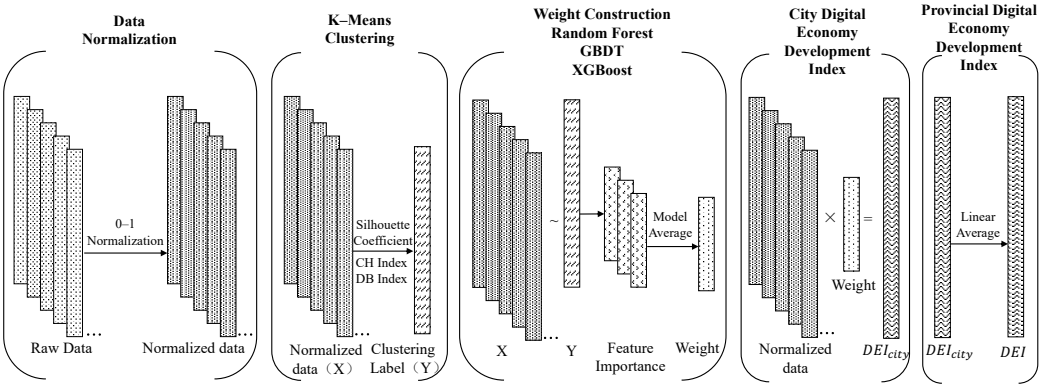


Figure 1. Flowchart of the digital economy development index calculation method.

3.2.3. Measurement and Decomposition of Multidimensional Inequality in Digital Economy Development

This paper utilizes a multidimensional Gini coefficient to measure the multidimensional inequality of digital economy development within provinces and further decomposes it to explore the independent contributions of each dimension. This section details the decomposition method used to analyze the influence of different dimensions.

The specific form of the multidimensional Gini coefficient is shown in Equation (5). This metric involves aggregating city-level data within each province as well as the heterogeneity across different dimensions of digital economy development, essentially representing a "double aggregation" approach (Decancq and Lugo [35]). For demonstration purposes, the disparities across the five dimensions of digital economy development analyzed in this paper are used as an example, with $m = 5$, to derive the multidimensional Gini coefficient.

$$MGini_i = 1 - \frac{\left[\sum_{k=1}^{5} w_{ik}\left(\sum_{c=1}^{n_i}\left[\left(\frac{r_{ck}}{n_i}\right)^{\gamma} - \left(\frac{r_{ck}-1}{n_i}\right)^{\gamma}\right]x_{ck}\right)^{\beta}\right]^{\frac{1}{\beta}}}{\left(\sum_{k=1}^{5} w_{ik}\mu_{ik}^{\beta}\right)^{\frac{1}{\beta}}} \quad (5)$$

In Equation (5), $MGini_i$ represents the multidimensional Gini coefficient for province i, measuring the extent of multidimensional inequality in digital economy development within that province. $w_i = (w_{i1}, w_{i2}, w_{i3}, w_{i4}, w_{i5})$ denotes the weights of the five dimensions of digital economy development for province i, satisfying $w_{i1} + w_{i2} + w_{i3} + w_{i4} + w_{i5} = 1$, and $w_{ik} > 0$. x_{ck} represents the development level of city c in province i along

the k-th dimension of digital economy development; r_{ck} is the rank of x_{ck} in the ordered sequence; n_i is the number of cities within province i; and $\gamma > 0$ is the inequality aversion parameter, set to 2 in this paper to ensure sufficient sensitivity to changes in inequality across the income distribution.

Although the multidimensional Gini coefficient achieves aggregation of both intra-group and inter-group information, there has been limited research on its decomposition. Drawing on the specific form of the multidimensional Gini coefficient, this paper uses the Kmenta approximation method (Kmenta [36]) applied to CES production functions, expanding it in a second-order Taylor series at $\beta = 0$ to achieve decomposition of the multidimensional Gini coefficient.

Let $\Phi_{ik} = \left(\sum_{c=1}^{n_i} \left[\left(\frac{r_{ck}}{n_i} \right)^{\delta} - \left(\frac{r_{ck}-1}{n_i} \right)^{\delta} \right] x_{ck} \right)$, then Equation (5) can be rewritten as

$$MGini_i = 1 - \frac{\left(\sum_{k=1}^{5} w_{ik} \Phi_{ik}^{\beta} \right)^{\frac{1}{\beta}}}{\left(\sum_{k=1}^{5} w_{ik} \mu_{ik}^{\beta} \right)^{\frac{1}{\beta}}} \tag{6}$$

Let $\psi_i = \sum_{k=1}^{5} w_{ik} \mu_{ik}^{\beta}$, then Equation (6) can be further rewritten as

$$MGini_i = 1 - \left[\sum_{k=1}^{5} w_{ik} \left(\frac{\Phi_{ik}}{\psi_i^{\frac{1}{\beta}}} \right)^{\beta} \right]^{\frac{1}{\beta}} = 1 - \Phi_i \tag{7}$$

Let $\tilde{\Phi}_{ik} = \frac{\Phi_{ik}}{\psi_i^{\frac{1}{\beta}}}$. Taking the logarithm of both sides of $\Phi_i = \left[\sum_{k=1}^{5} w_{ik} \left(\tilde{\Phi}_{ik} \right)^{\beta} \right]^{\frac{1}{\beta}}$ and expanding in a second-order Taylor series around $\beta = 0$, the expression becomes

$$\ln \Phi_i = w_{i1} \ln \tilde{\Phi}_{i1} + w_{i2} \ln \tilde{\Phi}_{i2} + w_{i3} \ln \tilde{\Phi}_{i3} + w_{i4} \ln \tilde{\Phi}_{i4} + w_{i5} \ln \tilde{\Phi}_{i5} + \frac{\beta}{2} \kappa_i \tag{8}$$

In Equation (8), $\kappa_i = \sum_{k=1}^{4} \sum_{k'=k+1}^{5} w_{ik} w_{ik'} \left(\ln \frac{\tilde{\Phi}_{ik}}{\tilde{\Phi}_{ik'}} \right)^2$. Exponentiating both sides of Equation (8) yields

$$\Phi_i = \left(\tilde{\Phi}_{i1}^{w_{i1}} \right) \left(\tilde{\Phi}_{i2}^{w_{i2}} \right) \left(\tilde{\Phi}_{i3}^{w_{i3}} \right) \left(\tilde{\Phi}_{i4}^{w_{i4}} \right) \left(\tilde{\Phi}_{i5}^{w_{i5}} \right) e^{\frac{\beta}{2} \kappa_i} \tag{9}$$

Substituting Equation (9) into Equation (7), the expression becomes:

$$MGini_i = 1 - \left(\tilde{\Phi}_{i1}^{w_{i1}} \right) \left(\tilde{\Phi}_{i2}^{w_{i2}} \right) \left(\tilde{\Phi}_{i3}^{w_{i3}} \right) \left(\tilde{\Phi}_{i4}^{w_{i4}} \right) \left(\tilde{\Phi}_{i5}^{w_{i5}} \right) e^{\frac{\beta}{2} \kappa_i} \tag{10}$$

Referring to Naga and Geoffard [37] and Hufe et al. [38], this paper adopts a linear approximation of a multivariate function. The standardization reference point, where all decomposition dimensions are in a state of complete equilibrium (i.e., $\tilde{\Phi}_{i1} = \tilde{\Phi}_{i2} = \tilde{\Phi}_{i3} = \tilde{\Phi}_{i4} = \tilde{\Phi}_{i5} = 1$), is expanded to derive the linear decomposition form in Equation (10), as shown in Equation (11). This decomposition allows the individual components of the multidimensional inequality index to measure the marginal contribution of each dimension to the overall inequality.

$$\begin{aligned} MGini_i &= \Phi_i|_{\tilde{\Phi}_{i1}=\tilde{\Phi}_{i2}=\tilde{\Phi}_{i3}=\tilde{\Phi}_{i4}=\tilde{\Phi}_{i5}=1} + \sum_{k=1}^{5} \left. \frac{\partial \Phi_i}{\partial \tilde{\Phi}_{ik}} \right|_{\tilde{\Phi}_{i1}=\tilde{\Phi}_{i2}=\tilde{\Phi}_{i3}=\tilde{\Phi}_{i4}=\tilde{\Phi}_{i5}=1} (\tilde{\Phi}_{ik} - 1) \\ &= w_{i1}(1 - \tilde{\Phi}_{i1}) + w_{i2}(1 - \tilde{\Phi}_{i2}) + w_{i3}(1 - \tilde{\Phi}_{i3}) + w_{i4}(1 - \tilde{\Phi}_{i4}) + w_{i5}(1 - \tilde{\Phi}_{i5}) \end{aligned} \tag{11}$$

Due to the varying degrees of multidimensional inequality in digital economy development across provinces, directly using the five terms on the right-hand side of Equation (11) as inequality indices for each decomposition dimension does not provide a consistent

upper limit, thereby lacking comparability across provinces. This paper normalizes the decomposition by dividing both sides of Equation (11) by $MGini_i$, resulting in a standardized decomposition as shown in Equation (12). This decomposition ensures that the value of each component is within the range $[0, 1]$ and that the sum of the contributions across the five dimensions equals 1, thereby achieving consistency and comparability.

$$\begin{aligned} 1 &= \frac{w_{i1}(1-\tilde{\Phi}_{i1})}{MGini_i} + \frac{w_{i2}(1-\tilde{\Phi}_{i2})}{MGini_i} + \frac{w_{i3}(1-\tilde{\Phi}_{i3})}{MGini_i} + \frac{w_{i4}(1-\tilde{\Phi}_{i4})}{MGini_i} + \frac{w_{i5}(1-\tilde{\Phi}_{i5})}{MGini_i} \\ &= MGini_{i1} + MGini_{i2} + MGini_{i3} + MGini_{i4} + MGini_{i5} \end{aligned} \tag{12}$$

In Equation (12), the five terms on the right side represent the relative contributions of the five dimensions to the multidimensional inequality in digital economy development in the i-th province: internet penetration ($MGini_{i1}$), information technology talent ($MGini_{i2}$), communication service development ($MGini_{i3}$), mobile communication penetration ($MGini_{i4}$), and digital transaction development ($MGini_{i5}$). It is worth noting that this paper uses an approximate decomposition method for the multidimensional Gini coefficient. Compared with traditional decomposition methods for inequality indices—such as Dagum decomposition or the multidimensional AKS1 index decomposition, which include interaction terms—this approach removes the interaction terms in the decomposition of the multidimensional Gini coefficient, isolating each dimension's independent contribution to overall inequality. This treatment avoids the confounding effects introduced by interaction terms, providing a clearer and more direct view of each dimension's impact on overall inequality, thereby enabling a more accurate quantification of each dimension's independent contribution to the overall disparity.

4. Empirical Results

4.1. Baseline Regression Results

The baseline regression results for the impact of digital economy development on regional income gaps are presented in Table 1. These results show a clear U-shaped relationship between digital economy development and intra-provincial regional income gaps, supporting Hypothesis H1. This suggests that while regional income gaps decrease during the early stages of digital economy development, they may expand as digital economy development progresses beyond a certain threshold.

As shown in Table 1, the coefficients for the first-order term of the digital economy index (DEI) in Models (1) and (2) are -0.300 and -0.318, respectively, both significantly negative at the 5% level. This finding indicates that the spread of digital technologies and the construction of digital infrastructure reduce regional income gaps within provinces, particularly in less-developed regions where digitalization offers more opportunities for market participation and resource allocation.

However, the coefficients for the second-order term of the digital economy index ($DEI2$) in Models (1) and (2) are 0.466 and 0.444, respectively, both significantly positive at the 1% level. This further supports the existence of a U-shaped relationship, suggesting that as digital economy development continues, economic resources become increasingly concentrated in regions with a digital advantage. These digitally advanced regions benefit more from a digital economy, widening income gaps with less-developed areas. In the early stages of digital economy development, efficiency gains are more evenly distributed across regions, but as digital technologies mature, more advanced regions are better positioned to attract capital, technology, and skilled labor, leading to a widening income gap.

The findings of this baseline model confirm the U-shaped relationship between digital economy development and regional income gaps. In the initial stages, digital economy development helps reduce income disparities, but as development continues, regions with a digital advantage accumulate more benefits, exacerbating the income gap with other regions. These results highlight the dynamic and complex impact of a digital economy on income distribution.

Table 1. Baseline model regression results: the impact of digital economy development on regional income gaps.

Variables	(1) Gini	(2) Gini
DEI	−0.300 **	−0.318 **
	(−2.268)	(−2.530)
DEI2	0.466 ***	0.444 ***
	(3.808)	(3.675)
gov		0.041
		(0.331)
rd		−1.172
		(−0.507)
stru		−0.006
		(−1.049)
tech		−0.983 **
		(−2.687)
gdpp		−0.000
		(−0.316)
Constant	0.841 ***	0.894 ***
	(26.660)	(11.025)
Observations	330	330
R-squared	0.972	0.974
Province	YES	YES
Year	YES	YES

Note: ***, $p < 0.01$; **, $p < 0.05$; *, $p < 0.1$; robust t-statistics in parentheses.

The U-shaped relationship is further illustrated in Figure 2, where the red line represents the range of the digital economy index for each province from 2011 to 2021, with a minimum value of 0.146 and a maximum value of 0.835. The green line marks the turning point of the U-shaped relationship, which is calculated to be 0.358 based on the baseline model regression results. This figure visually reinforces the dynamic nature of the relationship between digital economy development and regional income gaps. Notably, the digital economy index of most provinces in China lies to the left of this turning point, meaning that for these provinces, the development of the digital economy is associated with a reduction in regional income gaps. However, for a few more-developed provinces, such as Shanghai, where the average digital economy index is 0.703, the index has surpassed the turning point. In these provinces, further digital economy development may exacerbate regional income gaps, as more-developed regions with greater digital resources are positioned to benefit disproportionately, leaving less-developed areas behind.

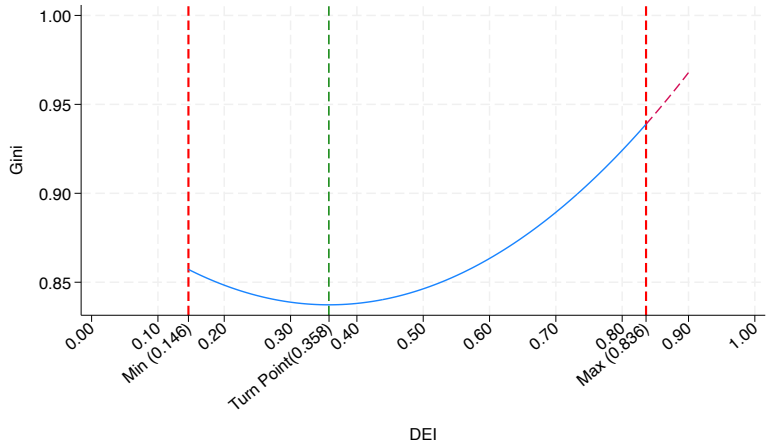


Figure 2. U-shaped relationship between digital economy development and regional income gaps. Note: The red line represents the range of the digital economy index for each province from 2011 to 2021, with a minimum value of 0.146 and a maximum value of 0.835. The green line indicates the turning point of the U-shaped relationship, which is calculated to be 0.358 based on the baseline model regression results.

The practical significance of these findings is crucial for policymakers aiming to foster balanced regional development. As digital economy development continues to expand, it is vital to ensure that resources are distributed equitably across regions. Policymakers should be cautious of the potential for excessive concentration of economic resources in digitally advanced regions, which could lead to widening income gaps. A more balanced approach is needed to ensure that less-developed regions can also benefit from the digital economy's growth.

4.2. Endogeneity and Robustness Tests

The baseline regression analysis indicates that digital economy development reduces regional income gaps but also exhibits a significant U-shaped nonlinear relationship. Considering that the baseline model's regression results may face various estimation issues, this paper conducts the following endogeneity and robustness tests to address potential issues related to variable selection and endogeneity in the baseline model.

4.2.1. Using Instrumental Variables to Mitigate Endogeneity

The baseline model controls for provincial and year fixed effects, which helps to alleviate endogeneity issues stemming from omitted explanatory variables to some extent. However, a potential reverse causality may exist between digital economy development and regional income gaps. Specifically, variations in income distribution could influence the development of a digital economy. For instance, in provinces with higher income levels or smaller income disparities, governments and enterprises are more capable of allocating resources toward digital infrastructure and adopting advanced digital technologies. In such contexts, residents' stronger purchasing power accelerates digital economy development. Advantages in technology, talent, and capital in these areas further facilitate the diffusion and application of digital innovations, promoting the expansion of a digital economy. Conversely, in provinces with lower income levels or wider income disparities, a lack of funding and technological capabilities may hinder the establishment of digital infrastructure, and residents' demand for digital products and services is relatively weak, resulting in slower progress in digital economy development. Consequently, income levels and disparities in these provinces may exert reverse causality on digital economy growth, introducing biases in OLS estimation and limiting the ability to accurately identify the causal relationship between a digital economy and regional income gaps. To address this issue, this paper employs the instrumental variable approach to mitigate endogeneity concerns, ensuring the reliability of the regression results.

In constructing the instrumental variable, this paper follows the approach of Yang and Jiang [39] and Zhao et al. [32], selecting the product of the number of fixed-line telephones per 100 people in each province in 1984 and total fixed asset investment in the information transmission, software, and information technology service sectors as the instrumental variable for the digital economy development level (DEI). The rationale for choosing the 1984 fixed-line telephone penetration rate lies in its relevance to early internet access, which primarily relied on dial-up technology through telephone lines. Provinces with higher telephone penetration were better positioned to establish an information network infrastructure, which in turn influenced the later adoption of internet technologies. This link establishes the instrumental variable's relevance to digital economy development. Additionally, fixed-line telephones, as a traditional communication tool, no longer exert a significant direct impact on economic development, especially with the decline in their usage over time, thus satisfying the exclusion restriction necessary for a valid instrument. This ensures that the fixed-line penetration rate influences digital economy development without directly affecting income gaps. To further mitigate potential price-level biases, fixed asset investment in the information technology service industry is deflated using the fixed asset investment price index, with the base year set to 2000. This adjustment provides a more accurate representation of actual investment levels over time, enhancing both the economic significance and the practical applicability of the constructed instrumental variable.

Models (1) and (2) in Table 2 report the results of the instrumental variable regression using two-stage least squares estimation (2SLS). These models are designed to address potential endogeneity issues, particularly reverse causality between digital economy development and regional income gaps. Following Du et al. [40], the baseline model is extended with quadratic terms to test the validity of the instrumental variables used. Model (1) presents the regression results of the instrumental variable iv on DEI, and Model (2) shows the results for the quadratic term $iv2$ as an instrument for $DEI2$. By incorporating these quadratic terms, this paper ensures that the instruments remain valid for capturing the nonlinear relationship between digital economy development and regional income gaps.

Table 2. Regression results of endogeneity and robustness tests.

Variables	(1) Gini	(2) Gini	(3) Theil	(4) Gini	(5) Gini	(6) Gini	(7) Gini
DEI	−3.639 **	−1.464 ***	−0.561 **				
	(−2.215)	(−2.964)	(−2.694)				
DEI2	3.103 **	1.618 ***	0.669 ***				
	(2.354)	(3.219)	(2.808)				
ces_DEI				−0.169 **			
				(−2.123)			
ces_DEI2				0.331 ***			
				(3.410)			
rf_DEI					−0.311 **		
					(−2.550)		
rf_DEI2					0.392 ***		
					(3.526)		
GBDT_DEI						−0.312 **	
						(−2.639)	
GBDT_DEI2						0.366 ***	
						(3.386)	
XGBoost_DEI							−0.179 *
							(−1.800)
XGBoost_DEI2							0.385 **
							(2.580)
gov	−0.118	0.017	−0.299	0.024	0.045	0.033	0.045
	(−0.607)	(0.166)	(−1.290)	(0.190)	(0.345)	(0.249)	(0.361)
rd	−4.031	−0.541	−1.655	−1.070	−1.282	−1.463	−1.227
	(−1.409)	(−0.366)	(−0.525)	(−0.468)	(−0.543)	(−0.619)	(−0.523)
stru	−0.014	−0.008	−0.001	−0.006	−0.006	−0.005	−0.007
	(−1.428)	(−1.607)	(−0.061)	(−1.089)	(−1.009)	(−0.965)	(−1.160)
tech	−1.235 **	−0.901 ***	1.660 **	−1.030 **	−1.017 **	−1.010 **	−0.957 **
	(−2.305)	(−2.941)	(2.568)	(−2.633)	(−2.687)	(−2.586)	(−2.722)
gdpp	0.000	0.000	0.000	−0.000	−0.000	−0.000	−0.000
	(1.442)	(1.210)	(0.699)	(−0.743)	(−0.306)	(−0.460)	(−0.383)
Constant	1.637 ***	1.162 ***	0.579 ***	0.877 ***	0.900 ***	0.915 ***	0.857 ***
	(4.976)	(13.124)	(3.939)	(11.642)	(10.517)	(10.648)	(11.131)
Observations	330	330	330	330	330	330	330
R-squared	0.883	0.960	0.855	0.974	0.974	0.974	0.973
Province	YES	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES	YES

Note: ***, $p < 0.01$; **, $p < 0.05$; *, $p < 0.1$; robust t-statistics in parentheses. This table presents the results of various tests addressing endogeneity and robustness in analyzing the impact of digital economy development on regional income gaps. Model (1) conducts an instrumental variable regression on the digital economy index (DEI), while Model (2) performs the same regression for the squared term of DEI (DEI2). Model (3) replaces the Gini coefficient with the Theil index as a measure of regional income gaps. Model (4) modifies the aggregation method of the index to assess the robustness of the results. Models (5) to (7) apply machine learning-based methods—Random Forest (RF), Gradient Boosting Decision Trees (GBDT), and eXtreme Gradient Boosting (XGBoost)—to directly calculate the weight of the digital economy index based on feature importance. All models control for provincial and year fixed effects.

To confirm the validity of the instrumental variables, under-identification and weak instrument tests are conducted for both *iv* and *iv*2. The results significantly reject the null hypotheses of "under-identification" and "weak instruments", providing robust evidence that the instrumental variables are relevant and valid for addressing endogeneity. These tests ensure that the instruments are not only correlated with the endogenous regressor (digital economy development) but also satisfy the exclusion restriction, meaning that they influence regional income gaps only through digital economy development and not directly.

In Model (1), the regression coefficient of DEI is -3.639, and that of $DEI2$ is 3.103, both significant at the 5% level. In Model (2), the coefficients for DEI and $DEI2$ are -1.464 and 1.618, respectively, both significant at the 1% level. These results indicate that even after addressing potential endogeneity issues, the U-shaped nonlinear relationship between digital economy development and regional income gaps remains significant. This reinforces the main conclusions of the baseline model, specifically that digital economy development initially reduces regional income gaps but may lead to widening gaps at later stages as more-developed regions accumulate digital advantages.

By performing these robustness tests, this paper ensures that the observed U-shaped relationship is not driven by omitted variables or reverse causality, but reflects a genuine causal link between digital economy development and regional income gaps. This strengthens the reliability of the study's conclusions and underscores the importance of monitoring digital economy development to avoid exacerbating regional income gaps.

4.2.2. Alternative Measurement Method for Regional Income Gaps

To further verify the impact of digital economy development on regional income gaps, this paper uses the Theil Index as an alternative measure instead of the Gini coefficient. The Theil Index is an entropy-based measure that assesses inequality by calculating the dispersion in income or economic activity distribution. Compared with the Gini coefficient, the Theil Index offers greater flexibility in decomposing differences in income gaps within and between regions. The specific calculation method is as follows:

$$Theil_i = \sum_{c=1}^{n_i} \frac{y_c}{Y_i} \ln\left(\frac{y_c}{Y_i \mu_i}\right) \tag{13}$$

In Equation (13), $Theil_i$ represents the Theil Index for province i. Specifically, n_i is the number of cities within the province, y_c denotes the nighttime light intensity of city c (used as a proxy for income level), Y_i is the total nighttime light intensity of all cities in the province, and μ_i is the average nighttime light intensity of these cities. The Theil Index reflects inequality by calculating the logarithmic deviation of each region's share of light intensity, with values ranging from zero to positive infinity. Higher values of the index indicate greater income gaps.

In Model (3) of Table 2, the regression results using the Theil Index as the dependent variable show that the regression coefficient for DEI is -0.561, significant at the 5% level, and the coefficient for $DEI2$ is 0.669, significant at the 1% level. These coefficient results align with those based on the Gini coefficient and continue to support the U-shaped nonlinear relationship between digital economy development and regional income gaps. Specifically, digital economy development reduces income gaps in its early stages but may increase gaps at later stages. Overall, the regression results remain significant after substituting the income gap measure, confirming that the baseline model's findings are robust across different inequality metrics and further supporting the nonlinear impact of digital economy development on regional income gaps.

4.2.3. Alternative Aggregation Method for the Digital Economy Development Index

To further validate the robustness of the digital economy development measurement, this paper replaces the aggregation method for calculating each province's digital economy development index used in the baseline model. The baseline model calculates the provincial

index as the simple arithmetic mean of each city's digital economy index. In this subsection, a Constant Elasticity of Substitution (CES) function is utilized to aggregate the digital economy indices of individual cities, aiming to capture the complementarity between cities within each province in the context of digital economy development more accurately. The elasticity of substitution in the CES function is set to 0.5, indicating a certain degree of complementarity among cities' development levels within each province rather than a simple average. The calculation formula for the CES index is as follows:

$$ces_DEI_i = \left[\sum_{c=1}^{n_i} \left(\frac{DEI_c^{(\delta-1)/\delta}}{n_i} \right) \right]^{\delta/(\delta-1)} \tag{14}$$

where ces_DEI_i represents the digital economy development index of province i, aggregated using the CES function, to measure the overall level of digital economy development in the province. Specifically, n_i is the number of cities in the province, DEI_c is the digital economy index of city c in province i, and δ is the elasticity of substitution parameter, set to 0.5 in this analysis to capture the complementarity among cities' digital economy development levels.

Model (4) in Table 2 reports the regression results using the provincial digital economy index aggregated by the CES function. The regression coefficient for the digital economy development level ces_DEI is -0.169, significant at the 5% level, and the coefficient for ces_DEI2 is 0.331, significant at the 1% level. Consistent with the baseline model, this result supports a U-shaped relationship: in the early stages of digital economy development, increasing the digital economy level helps reduce regional income gaps, but as development reaches a certain level, income gaps may widen. In summary, even with an alternative method for aggregating the provincial digital economy index, the regression results in terms of coefficient signs and significance remain consistent with the baseline model, further verifying the robustness of the U-shaped nonlinear relationship between digital economy development and regional income gaps.

4.2.4. Alternative Weighting Methods for the Digital Economy Development Index

In Models (5) to (7), the robustness of the digital economy development index is further verified by independently applying weights calculated through three distinct methods—Random Forest (RF), GBDT, and XGBoost—to construct the provincial-level digital economy index, instead of using the arithmetic mean of the weights derived from these methods. Regression analyses are conducted using weights calculated separately by each of these machine learning methods, enabling an examination of whether the impact of a digital economy on regional income gaps remains consistent across different weighting approaches, thereby enhancing the robustness of the results.

The regression results in Table 2 indicate a consistently significant U-shaped nonlinear relationship between digital economy development and regional income gaps, regardless of the weighting method used. Specifically, in Model (5), using Random Forest weights, the regression coefficient for rf_DEI is -0.311, significant at the 5% level, while the coefficient for rf_DEI2 is 0.392, significant at the 1% level. In Model (6), using GBDT weights, the coefficient for $GBDT_DEI$ is -0.312, significant at the 5% level, and the coefficient for $GBDT_DEI2$ is 0.366, also significant at the 1% level. In Model (7), using XGBoost weights, the coefficient for $XGBoost_DEI$ is -0.179, significant at the 10% level, and the coefficient for $XGBoost_DEI2$ is 0.385, significant at the 5% level. These results indicate that, regardless of the machine learning method used to calculate weights, the direction and significance of the regression coefficients remain consistent, further demonstrating the robustness of the impact of digital economy development on regional income gaps across different weighting methods.

In summary, the endogeneity and robustness tests support the core hypothesis of this paper: that digital economy development initially reduces regional income gaps, but as

digital economy development deepens, income gaps may gradually widen, resulting in a significant U-shaped nonlinear relationship.

4.3. Heterogeneity Across Economic Zones

In the heterogeneity analysis, this paper examines the income distribution effects of digital economy development across different economic areas to further reveal its variability across spatial divisions. Specifically, the sample is divided into two subsamples based on economic zones: the Yangtze River Economic Belt (YEB) and non-Yangtze River Economic Belt (The Yangtze River Economic Belt (YEB) covers 11 provinces and cities, including Shanghai, Jiangsu, Zhejiang, Anhui, Jiangxi, Hubei, Hunan, Chongqing, Sichuan, Yunnan, and Guizhou, spanning an area of approximately 2.0523 million square kilometers, accounting for 21.4% of China's total landmass, with population and GDP each exceeding 40% of the national total.). Additionally, this paper examines the disparities between municipalities and nonmunicipal areas (The municipalities include Beijing, Tianjin, Shanghai, and Chongqing.). The regression results are shown in Table 3.

Table 3. Regression Results of Spatial Heterogeneity Analysis.

Variables	(1) YEB Gini	(2) Non-YEB Gini	(3) Municipalities Gini	(4) Non-Municipalities Gini
DEI	−0.734 **	−0.277 *	−0.160	−0.188 *
	(−2.885)	(−1.892)	(−0.488)	(−1.681)
DEI2	0.626 ***	0.467 ***	0.236	0.324 ***
	(4.614)	(3.012)	(0.999)	(2.865)
gov	−0.588 *	0.086	−0.676	0.098
	(−1.972)	(0.662)	(−1.668)	(0.948)
rd	5.439	−0.751	5.397 *	−2.033
	(1.051)	(−0.464)	(1.939)	(−1.510)
stru	0.002	−0.011 *	0.012	−0.007 *
	(0.272)	(−1.887)	(0.951)	(−1.685)
tech	1.027	−1.212 ***	0.088	−0.566 *
	(1.160)	(−3.909)	(0.116)	(−1.849)
gdpp	−0.000 *	−0.000	−0.000	−0.000
	(−2.096)	(−0.344)	(−0.275)	(−0.244)
Constant	1.030 ***	0.877 ***	0.508 ***	0.890 ***
	(9.478)	(12.240)	(3.766)	(14.157)
Observations	121	209	44	286
R-squared	0.988	0.965	0.994	0.958
Province	YES	YES	YES	YES
Year	YES	YES	YES	YES

Note: ***, $p < 0.01$; **, $p < 0.05$; *, $p < 0.1$; robust t-statistics in parentheses.

For the sample from the Yangtze River Economic Belt, the income distribution effects of digital economy development exhibit a pronounced U-shaped pattern. The regression coefficients show that the linear term of the digital economy development index is −0.734, significantly negative at the 5% level, while the quadratic term is 0.626, significantly positive at the 1% level. This indicates that in the Yangtze River Economic Belt, digital economy development initially significantly reduces regional income gaps, but as it continues to deepen, the income gap begins to widen. This may be because the economic zone has more advanced infrastructure and a stronger economic foundation; the dividends of digital economy development are more broadly shared in the early stages but may concentrate in economic core areas over time, leading to increased income gaps.

For provinces outside the Yangtze River Economic Belt, the regression results also display a similar U-shaped relationship. The coefficients for the linear and quadratic terms of digital economy development are −0.277 and 0.467, respectively, significant at the 10% and 1% levels. Although the effect is less intense than in the Yangtze River

Economic Belt, these areas also experience an income distribution effect that first narrows and then widens. Compared with the Yangtze River Economic Belt, these areas have a weaker digital economy foundation and slower diffusion of digital technologies. The initial impact on reducing income gaps is more pronounced, but as digital economy development progresses, resources and technology gradually concentrate in core areas, leading to a widening income gap.

In the sample of municipalities, the coefficients for both the linear and quadratic terms of the digital economy development index are not statistically significant, suggesting that the income distribution effects of digital economy development in municipalities are not pronounced. This may be because municipalities are relatively economically developed with a somewhat stable income gap, and digital economy development has not significantly impacted income distribution. In contrast, for nonmunicipal areas, digital economy development exhibits a significant U-shaped relationship. The linear term coefficient is −0.188, and the quadratic term coefficient is 0.324, significant at the 10% and 1% levels, respectively. This result indicates that in nonmunicipal areas, the initial development of a digital economy helps reduce regional income gaps, but as economic resources increasingly concentrate in faster-developing cities, the income gap widens again.

In summary, the heterogeneity analysis reveals that the impact of digital economy development on regional income gaps varies significantly across different economic zones and administrative divisions. In more economically advanced areas like the Yangtze River Economic Belt and nonmunicipal regions, digital economy development initially narrows regional income gaps. However, as it progresses, the benefits tend to concentrate in core areas, widening the income gap. This concentration is driven by better infrastructure, higher investment in technology, and a more skilled workforce. In contrast, less-developed regions face challenges such as weaker infrastructure, lower digital literacy, and limited access to capital. While digital economy development initially helps reduce income gaps, these regions struggle to sustain benefits as digital technologies mature, exacerbating inequality.

To address these disparities, policymakers should focus on region-specific strategies. In advanced regions, the priority should be to ensure that digital economy benefits are more evenly distributed by investing in digital infrastructure and promoting digital skills in peripheral areas. In less-developed regions, policies should focus on improving digital infrastructure, boosting digital literacy, and supporting digital startups. Targeted interventions, such as subsidies for rural digital infrastructure and tech talent development, are essential to reduce the digital divide and ensure more inclusive growth. By addressing these regional differences, policymakers can foster more balanced and sustainable reductions in regional income gaps.

5. Further Analysis: Threshold Effect of Multidimensional Inequality in Digital Economy Development

In the baseline model analysis, this paper has confirmed the nonlinear impact of digital economy development on regional income gaps. However, digital economy development is characterized by significant multidimensional inequalities, where both the overall multidimensional inequality within provinces and disparities across specific dimensions may moderate its effectiveness in narrowing regional income gaps. To further explore the mechanisms through which multidimensional inequality influences the impact of digital economy development on the regional income gaps, this section introduces the multidimensional inequality index and its decomposed dimensions as threshold variables, testing the interaction effects between these factors and the level of digital economy development on reducing regional income gaps.

5.1. Multidimensional Inequality and Decomposition of Digital Economy Development

Figure 3 illustrates the trends of the multidimensional inequality index (*MGini*) for digital economy development, along with its five decomposed dimensions across provinces

(Due to the use of city-level data, detailed information on the districts under the municipalities is unavailable. Therefore, the analysis excludes the four municipalities: Beijing, Tianjin, Shanghai, and Chongqing.). The results highlight that the *MGini* index effectively captures the uneven distribution of digital economy development within provinces. In most regions, the multidimensional inequality index fluctuates, reflecting regional disparities in resource allocation during the digital economy's expansion. While some provinces are seeing trends toward greater balance, others continue to experience increasing inequality.

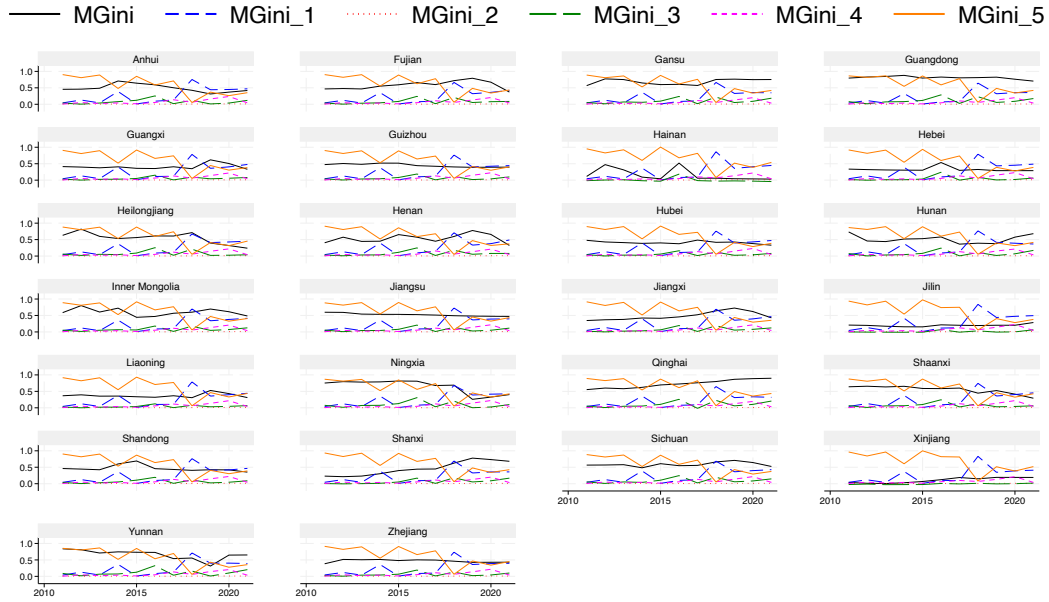


Figure 3. Multidimensional inequality index and decomposition results of digital economy development.

For example, provinces such as Zhejiang and Jiangsu show a year-on-year decline in the *MGini* index, indicating that resource distribution in these provinces is becoming more balanced as digital economy development progresses. As digital infrastructure, online transactions, and information technology improve, less-developed regions in these provinces are increasingly benefiting from the digital economy. On the other hand, provinces like Shanxi and Qinghai show an upward trend in their *MGini* indices, reflecting the concentration of digital economy resources in provincial capitals or more economically developed cities. As a result, less-developed areas are lagging in digital development, exacerbating inequalities within these provinces.

Focusing on the decomposed dimensions, the internet penetration dimension ($MGini_1$) and the digital transaction development dimension ($MGini_5$) are the primary contributors to multidimensional inequality in digital economy development across provinces. The internet penetration dimension ($MGini_1$) follows a "rise then fall" trend, indicating that in the early stages of digital economy development, internet resources were concentrated in specific regions, leading to greater inequality. However, as internet access expanded, the distribution of resources became more even, reducing inequality. In contrast, the digital transaction development dimension ($MGini_5$) shows a "fall then rise" trend, suggesting that while the early stages of digital transaction development saw relatively balanced resource allocation, the concentration of digital transactions in more-developed areas led to rising inequality. This dynamic pattern highlights that internet penetration and digital transactions are core drivers of multidimensional inequality in digital economy development. Furthermore, the information technology talent dimension ($MGini_2$) shows minimal variation and low levels of inequality across provinces, while the communication service development dimension ($MGini_3$) and mobile communication penetration dimension ($MGini_4$) demonstrate relatively stable or slightly declining trends.

This reveals significant regional disparities in the development of the digital economy, driven primarily by internet penetration and digital transactions. While some provinces benefit from a more equitable distribution of digital resources, others face widening gaps due to the concentration of digital economy resources in economically developed regions.

5.2. Threshold Effect Test Results

Table 4 presents the results of the threshold effect test using the multidimensional inequality index *Mgini* and its decomposed dimensions as threshold variables. The results show that *Mgini*, as a single threshold variable, has a significant threshold effect, with a threshold value of 0.5153 and an F-value of 31.04, significant at the 5% level (*p*-value = 0.013). This indicates that when the degree of inequality in digital economy development reaches this threshold, the impact of digital economy development on regional income gaps may change significantly. However, the double threshold test does not reach significance, with an F-value of 9.85 and a *p*-value of 0.337, suggesting that *Mgini* only exhibits a single threshold effect.

Table 4. Threshold Effect Test Result.

Threshold Variable		Threshold	F-Value	*p*-Value	1%	5%	10%
Mgini	Single Threshold	0.5153	31.04 **	0.013	31.209	17.043	14.614
	Double Threshold	0.5153 0.5365	9.85	0.337	35.219	24.261	17.867
$MGini_1$	Single Threshold	0.0124	42.52	0.260	54.115	50.249	46.821
	Double Threshold	0.0124 0.0448	13.77	0.820	39.762	32.359	28.515
$MGini_2$	Single Threshold	0.0014	42.52	0.193	53.801	48.295	46.313
	Double Threshold	0.0014 0.0101	15.32 ***	0.007	14.700	10.941	9.715
$MGini_3$	Single Threshold	0.1064	47.16	0.233	63.549	55.581	53.052
	Double Threshold	0.0388 0.1064	20.87 **	0.027	24.761	18.268	15.828
$MGini_4$	Single Threshold	0.1036	16.20	0.290	27.721	22.530	20.667
	Double Threshold	0.0129 0.0164	20.97 **	0.017	26.047	13.010	11.668
$MGini_5$	Single Threshold	0.4801	32.83 **	0.043	39.882	32.207	29.215
	Double Threshold	0.4801 0.7086	50.63 *	0.057	57.889	51.264	46.176

Note: ***, $p < 0.01$; **, $p < 0.05$; *, $p < 0.1$; the *p*-values and critical values are obtained by bootstrapping with 300 repetitions.

In terms of specific dimensions, the dimensions of information technology talent ($MGini_2$), communication service development ($MGini_3$), mobile communication penetration ($MGini_4$), and digital transaction development ($MGini_5$) all exhibit significant double threshold effects. Specifically, $MGini_2$ has double threshold values of 0.0014 and 0.0101, with an F-value of 15.32, significant at the 1% level (*p*-value = 0.007). For $MGini_3$, the double threshold values are 0.0388 and 0.1064, with an F-value of 20.87, significant at the 5% level (*p*-value = 0.027). The double threshold values for $MGini_4$ are 0.0129 and 0.0164, with an F-value of 20.97, also significant at the 5% level (*p*-value = 0.017). For $MGini_5$, the double threshold values are 0.4801 and 0.7086, with an F-value of 50.63, significant at the 10% level (*p*-value = 0.057). These results suggest that the impact of digital economy development on regional income gaps exhibits significant stratification across different inequality levels and that each developmental stage may have varying effects on narrowing income gaps. Additionally, the threshold effect of internet penetration ($MGini_1$) is not statistically significant, indicating that the gap in this dimension has a limited effect on the role of the digital economy in reducing the income gap across different levels of inequality.

In conclusion, the multidimensional inequality index *Mgini* exhibits a significant threshold effect, with each decomposed dimension showing distinct threshold characteristics. Specifically, the digital economy's impact on regional income gaps varies at different stages of inequality, with certain dimensions displaying notable double threshold effects. These findings highlight the importance of considering these thresholds when designing policy measures to address regional gaps. This issue will be further examined in the following section through the construction of threshold regression models.

5.3. Threshold Model Regression Results

To further explore the mechanisms of these threshold effects, this paper sets the different threshold values as dummy variables and interacts them with *DEI* to analyze the specific direction and intensity of the impact of digital economy development on regional income gaps under different levels of multidimensional inequality. Table 5 reports the results of the panel threshold test on the impact of digital economy development on regional income gaps. The following conclusions can be drawn from the regression results.

Table 5. Threshold Model Regression Results.

Variables	Mgini	(1) Gini	$Mgini_2$	(2) Gini	$Mgini_3$	(3) Gini	$Mgini_4$	(4) Gini	$Mgini_5$	(5) Gini
DEI	[0, 0.5153]	−0.174 ***	[0, 0.0014]	−0.060	[0, 0.0388]	−0.115 **	[0, 0.0129]	−0.203 ***	[0, 0.4801]	−0.113 **
		(−4.094)		(−1.288)		(−2.471)		(−5.015)		(−2.084)
DEI	(0.5153, 1]	−0.069	(0.0014,	−0.181 ***	(0.0388,	−0.047	(0.0129,	−0.064 *	(0.4801,	0.069 *
		(−1.407)	0.0101]	(−3.947)	0.1064]	(−0.905)	0.0164]	(−2.030)	0.7086]	(1.764)
DEI			(0.0101, 1]	−0.249 ***	(0.1064,1]	0.040	(0.0164, 1]	−0.197 ***	(0.7086, 1]	−0.062
				(−5.380)		(0.780)		(−5.124)		(−1.552)
gov		0.102		0.002		0.173 *		0.065		0.248 **
		(0.853)		(0.016)		(1.721)		(0.614)		(2.338)
rd		−5.238 **		−4.151 ***		−6.010 ***		−5.674 ***		−2.424
		(−2.697)		(−2.715)		(−3.645)		(−3.205)		(−1.134)
stru		−0.002		−0.005		−0.002		−0.006		−0.004
		(−0.297)		(−0.858)		(−0.342)		(−0.865)		(−0.558)
tech		−0.696		−0.591		−0.939		−0.930		−0.417
		(−1.152)		(−1.551)		(−1.589)		(−1.662)		(−0.942)
gdpp		−0.000		−0.000		−0.000		−0.000		−0.000
		(−1.136)		(−1.364)		(−0.865)		(−1.087)		(−0.910)
Constant		1.004 ***		1.031 ***		0.961 ***		1.037 ***		0.879 ***
		(11.449)		(14.109)		(11.111)		(12.635)		(10.719)
Observations		286		286		286		286		286
R-squared		0.232		0.296		0.315		0.246		0.355
r2_w		0.084		0.091		0.086		0.087		0.090
r2_b		0.039		0.037		0.037		0.038		0.036
r2_o		0.824		0.857		0.846		0.838		0.864

Note: ***, $p < 0.01$; **, $p < 0.05$; *, $p < 0.1$; robust t-statistics in parentheses. Model (1) uses the overall multidimensional inequality index (Mgini) as the threshold variable, assessing how changes in the level of inequality in digital economy development affect income gaps. Model (2) uses the inequality index for the information technology talent dimension ($Mgini_2$) as the threshold variable. Model (3) uses the inequality index for the communication service development dimension ($Mgini_3$) as the threshold variable. Model (4) employs the inequality index for mobile communication penetration ($Mgini_4$) as the threshold variable. Model (5) uses the inequality index for digital transaction development ($Mgini_5$) as the threshold variable.

5.3.1. Overall Multidimensional Inequality

The regression results of Model (1) set the multidimensional inequality index of digital economy development, *Mgini*, as the threshold variable, revealing a significant nonlinear impact of digital economy development on regional income gaps. When the *Mgini* index is in the lower range $[0, 0.5153]$, the coefficient of the digital economy development index (DEI) on regional income gaps ($Gini$) is -0.174, significantly negative at the 1% level. This suggests that in a context of relatively low multidimensional inequality, a more balanced distribution of resources and technology spillovers allows less-developed regions to more effectively benefit from the growth dividends of a digital economy, contributing to a reduction in regional income gaps. This effect may stem from the inclusive nature of the digital economy when inequality is low—characterized by the widespread application of

digital infrastructure and technology diffusion—enabling lagging regions to more equitably participate in market activities and improve productivity.

However, when the *Mgini* index exceeds 0.5153, the income distribution effect of digital economy development diminishes significantly, with the DEI coefficient dropping to -0.069 and losing significance. This shift indicates that under higher levels of multidimensional inequality, the impact of the digital economy on narrowing regional income gaps gradually fades. At this stage, resources and technology may be more concentrated in economically advanced areas, making it difficult for less-developed regions to absorb the benefits of digital economy development equally, and they may even face marginalization, thus limiting the balancing effect of the digital economy. High levels of inequality can restrict opportunities for less-developed regions within the digital economy, thereby weakening the role of the digital economy in reducing regional income gaps.

Overall, this result validates the theoretical hypothesis of this paper that there is a significant threshold effect in the impact of digital economy development on regional income gaps. When multidimensional inequality is at a lower level, digital economy development is more likely to promote balanced income distribution; however, under high inequality, the concentration of technology and resources may hinder the inclusivity of the digital economy, thereby limiting its positive influence on reducing regional income gaps. Next, the threshold effects of its decomposed dimensions are further examined and analyzed.

5.3.2. Information Technology Talent Dimension

In Model (2), the inequality index of the information technology talent dimension ($MGini_2$) is used as the threshold variable. When $MGini_2$ is in the lower range of $[0, 0.0014]$, the coefficient of the digital economy index (DEI) on regional income gaps is -0.060, and it is not significant, indicating that the impact of digital economy development on income gaps is limited at this stage.

However, when $MGini_2$ is in the range $(0.0014, 0.0101]$, the coefficient of DEI becomes -0.181, and it is significantly negative at the 1% level. This suggests that a moderate level of inequality in information technology talent helps to strengthen the impact of digital economy development on narrowing the income gap.

Furthermore, when $MGini_2$ exceeds 0.0101, the coefficient of DEI is -0.249, which remains significantly negative at the 1% level. This indicates that at higher levels of inequality, the effect of digital economy development on income gaps is the strongest. This result shows that inequality in information technology talent, within a certain range, can effectively promote the role of digital economy development in reducing regional income gaps, especially when inequality is moderate or high.

The reasons for this are twofold: first, when inequality is moderate or high, underdeveloped regions are more proactive in attracting technical talent and improving infrastructure to narrow the gap. This policy response enhances the expansion of the digital economy in these regions, thereby reducing income gaps. Second, at higher levels of inequality, technology and expertise from more-developed regions are more likely to spill over to underdeveloped areas, which, due to their lower base, have greater potential to absorb the benefits of digitalization. Therefore, inequality in information technology talent, at moderate to high levels, enhances the role of digital economy development in narrowing the income gap through policy interventions and spillover effects.

5.3.3. Communication Service Development Dimension

In the results of Model (3), using the inequality index of the communication service development dimension, $MGini_3$, the threshold variable reveals a double threshold effect of digital economy development on regional income gaps. When $MGini_3$ is below the threshold value of 0.0388, the coefficient of digital economy development on income gaps is -0.115 and is significant at the 5% level, indicating that in regions with relatively low inequality in communication services, digital economy development significantly reduces

income gaps. This may be due to the fact that, in a relatively balanced communication environment, less-developed regions can more easily obtain support for communication technology and resources, thereby facilitating these regions' absorption of digital economy benefits and narrowing the income gap.

However, when $MGini_3$ rises to a moderate inequality level, within the interval $(0.0388, 0.1064]$, the impact coefficient of digital economy development weakens to -0.047 and is no longer significant, indicating that the effect of digital economy development on reducing income gaps is significantly diminished at this level. In this range, the uneven distribution of communication resources has not yet triggered significant resource redistribution or policy intervention, making it difficult for less-developed areas to continually receive sufficient communication resource investment. Consequently, the income distribution effect of the digital economy is not prominent.

When the communication service inequality index $MGini_3$ exceeds the high threshold value of 0.1064, the coefficient of digital economy development turns positive at 0.040 but remains insignificant. This result suggests that in cases of high inequality in communication services, the positive effect of digital economy development on reducing income gaps is weakened and may even lead to an increase in income gaps. This situation may arise from the high concentration of communication resources in a few developed areas, which hinders the penetration of digital technology into less-developed regions and further exacerbates the digital divide.

In summary, the threshold effect of the communication service inequality index reveals its role in moderating the impact of digital economy development: when communication resources are relatively evenly distributed, the positive effect of digital economy development on reducing regional income gaps is most significant. However, as communication service inequality intensifies—particularly at high inequality levels—the positive impact of digital economy development on income gaps tends to weaken or even reverse. Therefore, improving the accessibility and investment of communication resources in less-developed regions and reducing regional inequality in communication services can help more broadly leverage the potential of digital economies in reducing regional income gaps.

5.3.4. Mobile Communication Penetration Dimension

The regression results of Model (4), which uses the inequality index of mobile communication penetration ($MGini_4$) as the threshold variable, reveal a significant double threshold effect of digital economy development on regional income gaps. When $MGini_4$ is in the low inequality range below 0.0129, the coefficient of the digital economy development index (DEI) is -0.203 and is significant at the 1% level, indicating that in regions with relatively balanced mobile communication penetration, digital economy development significantly reduces regional income gaps. This suggests that in a more equitable environment of communication penetration, digital resources are able to spread widely, allowing less-developed regions to effectively access and utilize these resources, thus fully sharing in the income growth opportunities brought by a digital economy and reducing regional income gaps.

When $MGini_4$ rises to a moderate inequality range of $(0.0129, 0.0164]$, the coefficient of digital economy development on income gaps weakens to -0.064, though it remains significant at the 10% level. This demonstrates the inhibitory effect of communication inequality on the equalizing role of the digital economy. Although digital economy development continues to reduce income gaps to some extent, moderate inequality restricts the flow of communication resources, thus weakening the equalizing effect of the digital economy.

When $MGini_4$ exceeds 0.0164, entering a high inequality range, the coefficient of digital economy development on income gap increases again to -0.197 and remains significant at the 1% level. This indicates that, even under high inequality, the digital economy still plays a significant role in reducing income gaps. The reason for this may be that under high inequality, less-developed regions typically receive stronger policy support, such as government subsidies and corporate infrastructure investment, helping them to

better integrate into a digital economy. Additionally, spillover effects from the mobile communication technology and expertise of developed regions to less-developed areas increase as the developed market approaches saturation, prompting companies to expand their technology and services to less-developed regions, thus providing opportunities to bridge the digital divide. Moreover, since less-developed regions have a lower initial resource base, the marginal benefits of newly introduced resources are more substantial, helping to raise local income levels.

In summary, the threshold effect of the mobile communication penetration inequality index indicates that digital economy development has the most pronounced impact on reducing income gaps at lower levels of inequality.

5.3.5. Digital Transaction Development Dimension

In Model (5), using the inequality index of digital transaction development, $MGini_5$, as the threshold variable, the threshold regression results reveal a significant double threshold effect of digital economy development on regional income gaps. When $MGini_5$ is in the low inequality range below 0.4801, the coefficient of the digital economy development index is -0.113 and is significant at the 5% level. This indicates that under relatively balanced digital transaction conditions, digital economy development effectively reduces regional income gaps. This effect may occur because, in a low-inequality environment, digital transaction development is more widespread, allowing regions to gain equitable access to the benefits of the digital economy, thereby promoting income equalization to some extent.

When $MGini_5$ rises to a moderate inequality range (0.4801, 0.7086], the coefficient of digital economy development turns positive, reaching 0.069, and is significant at the 10% level. This result suggests that at moderate levels of digital transaction inequality, the equalizing effect of digital economy development on income gaps may be suppressed, even exhibiting a gap-widening effect. This phenomenon may be attributed to certain developed regions accelerating their absorption of digital transaction dividends, further consolidating their advantage in the digital economy and placing less-developed regions at a greater disadvantage in terms of resource and technology flows, thereby exacerbating regional income gaps.

Further analysis shows that when $MGini_5$ exceeds 0.7086, entering a high inequality range, the coefficient of digital economy development again turns negative but is no longer significant. This indicates that in an environment of extremely high inequality in digital transactions, the positive effect of digital economy development on reducing income gaps becomes nonsignificant. A likely reason for this is that excessive inequality leads to a high concentration of resources and technology among regions, reducing the efficiency of digital economy dividends and limiting the gains less-developed regions can achieve from digital transactions. Extreme inequality may also result in fragmented regional economic development, hindering the broad-based sharing effects of the digital economy.

In summary, varying levels of inequality in digital transaction development have a significant moderating effect on the relationship between digital economy development and income gaps. At lower levels of inequality, digital economy development helps narrow the income gap; at moderate levels of inequality, however, it may exacerbate regional income gaps. In a context of high inequality, the positive effect of digital economy development on improving income gaps tends to weaken or even disappear. This result suggests that achieving balanced regional development in a digital economy requires an appropriate control of digital transaction inequality to prevent excessive concentration of resources and technology in a few regions, thereby maximizing the positive impact of a digital economy on narrowing income gaps.

The threshold effect analysis reveals that digital economy development has varying impacts on regional income gaps depending on the level of multidimensional inequality. In general, the positive effects of digital economy development on narrowing income gaps are most pronounced at lower levels of inequality across dimensions such as information technology talent, communication services, mobile penetration, and digital transaction

development. At higher levels of inequality, the effectiveness of a digital economy in reducing income gaps diminishes, and in some cases, may even exacerbate gaps. This emphasizes the need for policymakers to focus on managing inequality within specific dimensions to maximize the inclusive benefits of digital economy growth, ensuring that less-developed regions can effectively participate and benefit from digital transformation. The results confirm Hypothesis H2, which suggests that optimal levels of inequality can help digital economy development promote more balanced regional income distributions.

6. Conclusions

This paper explores the significant impact of digital economy development on income inequality in China. By exploring the intrinsic characteristics of a digital economy, it analyzes how digital economy development influences provincial regional income gaps through its underlying mechanisms and multidimensional threshold effects. This paper makes several contributions to the literature. First, it expands the theoretical and empirical understanding of how digital economy development affects regional income gaps, providing new insights into the complex relationship between technological development and income distribution. Second, the paper develops a measurement framework for digital economy development, using machine learning algorithms—such as Random Forest, GBDT, and XGBoost—to assess digital economy development at a more granular level using city-level data. Third, it advances methodological innovation by applying the Kmenta approximation for the decomposition of multidimensional inequality, isolating the independent contributions of each dimension, and offering a more nuanced understanding of how different facets of digital economy development contribute to regional income gaps.

The theoretical hypotheses are empirically tested using provincial panel data from China spanning 2011 to 2021. The main findings are as follows. First, the impact of digital economy development on regional income gaps within provinces follows a U-shaped relationship, initially narrowing and subsequently widening income gaps. This result remains robust even after controlling for various endogeneity issues and conducting robustness checks. Second, by applying the Kmenta approximation to decompose multidimensional inequality in digital economy development, this paper isolates each dimension's independent contribution to overall inequality by removing interaction terms, identifying internet penetration and digital transaction development as the primary drivers of multidimensional inequality in the digital economy. Third, by introducing the multidimensional inequality index of digital economy development and its components as threshold variables in the model, this paper reveals that moderate levels of multidimensional inequality in digital economy development significantly enhance its positive effect on reducing regional income gaps. However, when inequality surpasses this moderate range, the positive effect diminishes and may even intensify regional income gaps.

The findings of this paper offer several policy implications:

- Strengthening digital infrastructure in less-developed regions. Policy interventions should prioritize the expansion of digital infrastructure in less-developed regions, particularly in the early stages of digital economy development. Ensuring adequate access to digital infrastructure, such as internet connectivity and communication facilities, is essential for enabling more equitable participation in the digital economy and reducing income gaps.
- Balancing multidimensional inequality. Policymakers should emphasize coordinated development across different dimensions of the digital economy—such as information technology talent, digital transactions, and communication infrastructure—so as to avoid excessive disparities between regions. Balancing these dimensions will ensure that the benefits of digital economy development are more evenly distributed, thereby enhancing its positive impact on narrowing regional income gaps.
- Encouraging technology spillovers and cross-regional cooperation. The study underscores the importance of promoting technology spillovers and fostering collaboration between more digitally advanced regions and less-developed areas. By encouraging

resource sharing and technological cooperation, policymakers can help mitigate the risk of digital economy development exacerbating regional income gaps and ensure that the benefits of innovation are broadly distributed.

- Adapting policies to prevent widening inequality. As digital economy development matures, regional income gaps may widen due to the concentration of resources and technology in economically advanced areas. To address this issue, policies must be dynamically adjusted to prevent excessive resource concentration in a few regions, and to ensure that less-developed areas continue to benefit from digital economy development in a sustainable manner.

In addition to the contributions discussed, there are several limitations to this study that should be acknowledged. While the analysis of income gaps is based on city-level data, other crucial data, such as digital economy development indices, are only available at the provincial level due to data accessibility constraints. As a result, the study is unable to perform a more detailed analysis at the city or district level, which could provide further insights into the spatial variations and finer mechanisms behind regional income disparities. Future research could focus on expanding data coverage to lower administrative levels, enabling a more granular exploration of the relationship between digital economy development and regional income gaps. Furthermore, a more in-depth investigation of the dynamic effects of digital economy policies and their long-term impact on regional income gaps would enhance the understanding of digital economy development's role in shaping economic inequality.

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